

Project Guideline

1. Project – Machine Learning Application

This project is dedicated to implementing a **complete machine learning solution** to a problem of your choice.

You may select any real-world or synthetic dataset, provided that:

1. The problem is meaningful and non-trivial
2. It requires proper data preprocessing
3. It involves building, training, and evaluating a machine learning model
4. Your results are clearly analyzed

Examples of acceptable project types include:

- Classification (binary or multi-class)
- Regression
- Clustering
- Anomaly detection
- Recommender systems
- Time-series forecasting

2. Free Topic – ML Algorithm Study

Each team must choose **one machine learning algorithm that was NOT covered in class** and provide a clear, technical explanation and presentation.

Your Free Topic submission must include:

1. Theoretical explanation of the algorithm
2. Mathematical formulation (if applicable)
3. A simple example demonstrating how it works
4. Advantages, limitations, and typical applications

This section is strictly theory-oriented and does not require full implementation unless desired.

➤ Team Structure

- Each team may consist of **3-4 students**.
- A **Task Assignment Page** must be included in all submissions, detailing:
 - Tasks performed
 - Contribution percentage for each student
 - Signatures of all team members

Failure to include this page results in a **2% deduction**.

➤ **Activities & Deadlines**

1. **Project Report & Poster**

Due date: Announced Later

Each team must submit:

- A **typed and bound hardcopy** of the project report
- An **electronic zip file** including:
 - Source code
 - Dataset (or link if too large)
 - Sample outputs
 - PDF of the report
- An **A4 project poster** summarizing problem, method, and results.

2. **Presentations**

Discussions for **Project** and **Free Topic** will be scheduled later (time and room to be announced).

➤ **A Cover Page for the report which includes:**

- Project Title
- Course Number
- Student names, IDs, sections
- Date submitted

➤ **The Project Report Sections may include:**

1. Cover Page
2. Problem Description
3. Data Preparation
4. Methods and Models
5. Results and Evaluation
6. Discussion
7. Appendix
8. Task Assignment Page